Supplementary Materials for "Scalable Enumeration of Pareto-optimal Polymers for Computing Equilibrium Concentrations"

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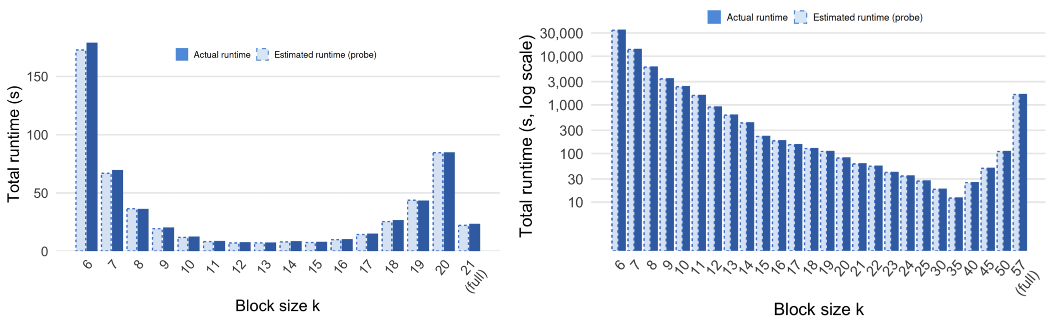
1 Implementation Details

As discussed in Section 6.2 of the main text, Figures 1a and 1b illustrate the runtime as a function of k for the Scaffolded DNA and AND gate linear cascade systems (Section 7.1 of main text), respectively. As observed, the estimated runtime is very close to the actual runtime, and there is an optimal value of k that minimizes the runtime.



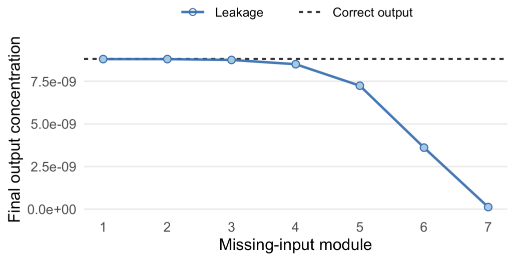
(a) Scaffolded DNA system ($|M| = 21$, $t = 5$).

(b) AND gate linear cascade, 7 modules ($|M| = 57$, $t = 5$).

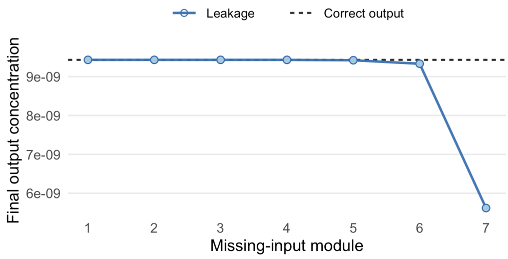
Figure 1 Probe-estimated total Hilbert basis runtime as a function of block size k, with support bound fixed at $t = 5$. Each point is the product of the per-block trial time (100-iteration probe) and the number of blocks in the covering design. The minimum of each curve identifies the selected k.

2 Leakage-Scaling Plots

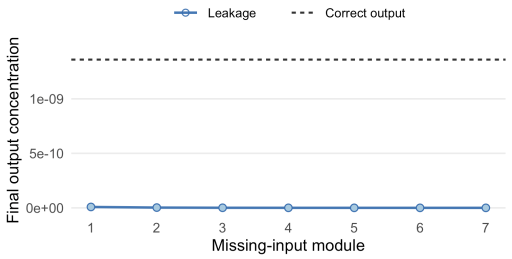
Section 7.4 of the main text analyzes leakage depending on which module has its input removed in a fixed-length linear cascade system. Here are the figures for this experiment across different values of E and concentration regimes A and B.



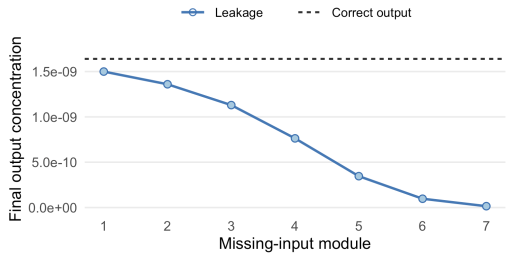
(a) Regime A, $E = -20\,\mathrm{kcal/mol}$



(b) Regime A, $E = -10\,\mathrm{kcal/mol}$



(c) Regime B, $E = -20\,\mathrm{kcal/mol}$

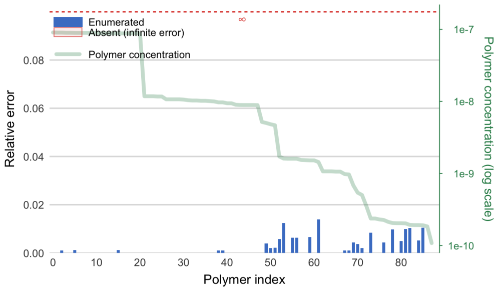


(d) Regime B, $E = -10\,\mathrm{kcal/mol}$

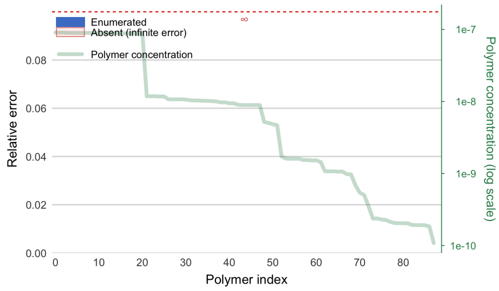
Figure 2 Final output concentration under single-input deletion in a seven-module linear cascade. The x-axis indicates the module whose input is removed. The solid curve shows the leakage concentration, and the dashed horizontal line shows the correct-output concentration when all inputs are present. The four panels compare Regimes A and B under bond energies $E = -20\,, -10\,\mathrm{kcal/mol}$.

3 Equilibrium-Accuracy Plots

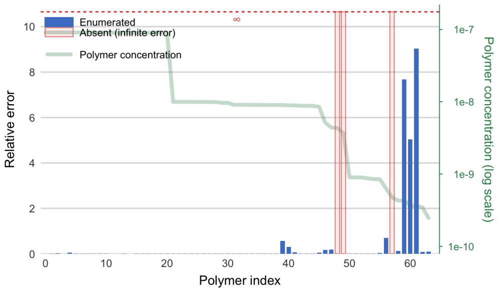
In Section 7.3 of the main text, we report the Regime B leakage condition with bond energy $E = -20\,\mathrm{kcal/mol}$. Here we include the corresponding concentration-error plots for both concentration regimes (A and B), each tested with different bond energies ($E = -20, -10\,\mathrm{kcal/mol}$). As shown in the figure, $t = 5$ provides a slightly more accurate equilibrium analysis compared to $t = 3$, although the concentrations of most equilibrium-relevant polymers are accurately captured even with the smaller support bound. Runtime for different cascade lengths and values of t is compared in Figure 4. Although the runtime for the covering-design algorithm still scales exponentially with the number of modules, the rate of growth is much lower than that of the full Hilbert basis computation.



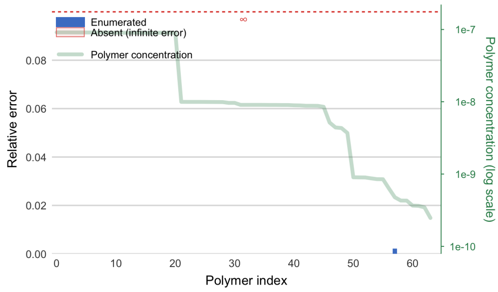
(a) Regime A leakage, $E = -10$ kcal/mol, $t = 3$.



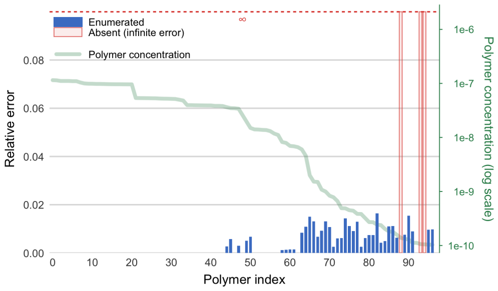
(b) Regime A leakage, $E = -10$ kcal/mol, $t = 5$.



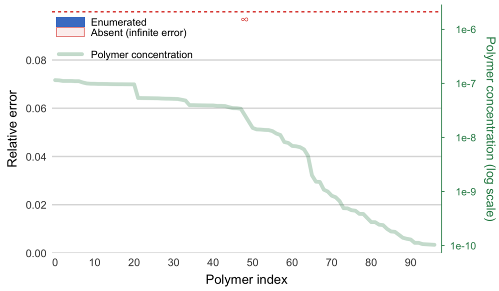
(c) Regime A leakage, $E = -20$ kcal/mol, $t = 3$.



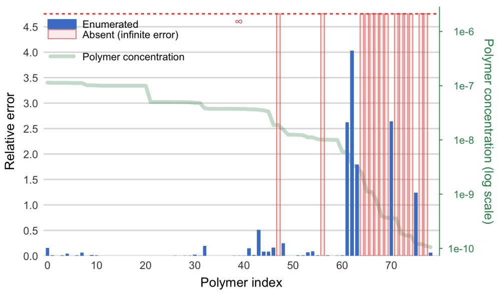
(d) Regime A leakage, $E = -20$ kcal/mol, $t = 5$.



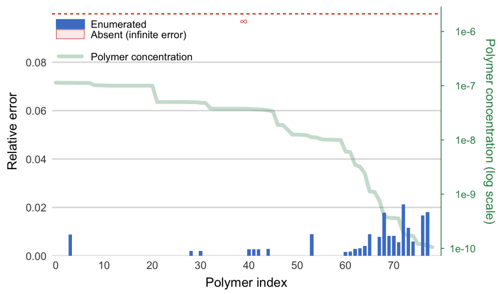
(e) Regime B leakage, $E = -10$ kcal/mol, $t = 3$.



(f) Regime B leakage, $E = -10$ kcal/mol, $t = 5$.



(g) Regime B leakage, $E = -20$ kcal/mol, $t = 3$.



(h) Regime B leakage, $E = -20$ kcal/mol, $t = 5$.

Figure 3 Relative concentration error for equilibrium-relevant polymers in the seven-module linear cascade under Regimes A and B with bond energies $E = -20, -10$ kcal/mol, mirroring saturated and unsaturated systems, respectively.

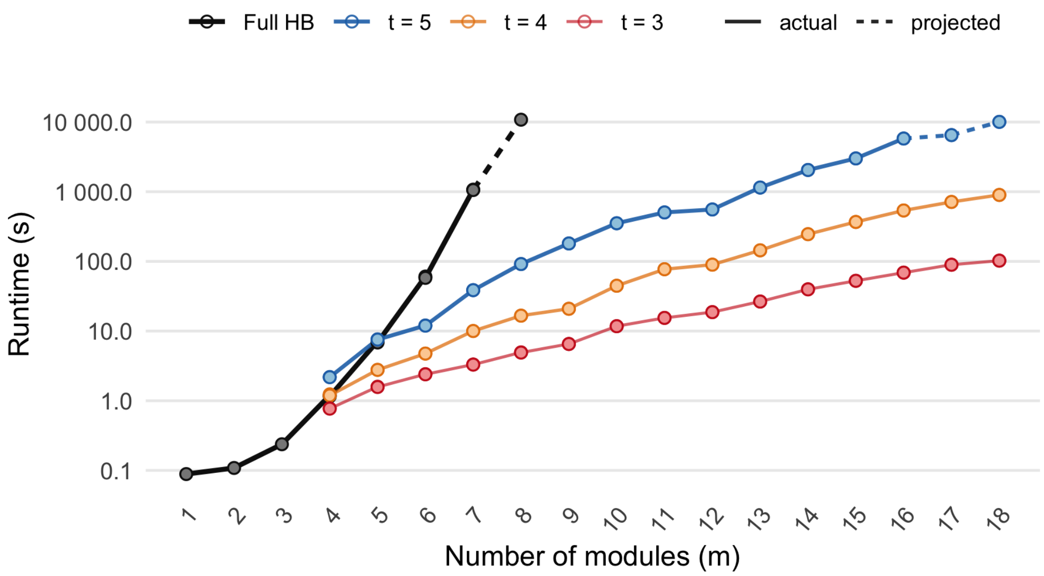


Figure 4 Runtime comparison for leakage experiments across linear cascade lengths and different parameters of the covering-design algorithm. Full HB represents the runtime required for full Hilbert basis computation using Normaliz, and the projected runtime is computed from the probing step of Section 6.2 of the main text.